

Network Traffic

http						
No.	Time	Source	Destination	Protocol	Length	Info
→ 1156	25.512991	192.168.68.114	103.230.106.216	HTTP	408	GET /time.php HTTP/1.1
← 1158	25.524797	103.230.106.216	192.168.68.114	HTTP	453	HTTP/1.1 200 OK (text/html)
3200	85.370579	192.168.68.114	103.230.106.216	HTTP	408	GET /time.php HTTP/1.1
3202	85.376741	103.230.106.216	192.168.68.114	HTTP	453	HTTP/1.1 200 OK (text/html)
7175	145.366362	192.168.68.114	103.230.106.216	HTTP	408	GET /time.php HTTP/1.1
7177	145.378247	103.230.106.216	192.168.68.114	HTTP	453	HTTP/1.1 200 OK (text/html)
9843	188.682255	192.168.68.114	103.230.106.216	HTTP	489	GET / HTTP/1.1
9852	188.691635	103.230.106.216	192.168.68.114	HTTP	1181	HTTP/1.1 200 OK (text/html)
9921	189.835567	192.168.68.114	103.230.106.216	HTTP	408	GET /time.php HTTP/1.1
9923	189.841660	103.230.106.216	192.168.68.114	HTTP	453	HTTP/1.1 200 OK (text/html)
9926	190.832561	192.168.68.114	103.230.106.216	HTTP	408	GET /time.php HTTP/1.1
9928	190.838552	103.230.106.216	192.168.68.114	HTTP	452	HTTP/1.1 200 OK (text/html)
9940	191.843724	192.168.68.114	103.230.106.216	HTTP	408	GET /time.php HTTP/1.1
9942	191.849480	103.230.106.216	192.168.68.114	HTTP	453	HTTP/1.1 200 OK (text/html)
9971	192.839513	192.168.68.114	103.230.106.216	HTTP	408	GET /time.php HTTP/1.1
9980	192.846708	103.230.106.216	192.168.68.114	HTTP	453	HTTP/1.1 200 OK (text/html)
10036	193.835808	192.168.68.114	103.230.106.216	HTTP	408	GET /time.php HTTP/1.1
10038	193.852312	103.230.106.216	192.168.68.114	HTTP	453	HTTP/1.1 200 OK (text/html)
10070	195.353328	192.168.68.114	103.230.106.216	HTTP	408	GET /time.php HTTP/1.1
10072	195.360749	103.230.106.216	192.168.68.114	HTTP	453	HTTP/1.1 200 OK (text/html)
10087	196.351804	192.168.68.114	103.230.106.216	HTTP	408	GET /time.php HTTP/1.1
10089	196.358427	103.230.106.216	192.168.68.114	HTTP	453	HTTP/1.1 200 OK (text/html)
10096	196.835745	192.168.68.114	103.230.106.216	HTTP	408	GET /time.php HTTP/1.1
10098	196.842006	103.230.106.216	192.168.68.114	HTTP	453	HTTP/1.1 200 OK (text/html)
10110	197.829575	192.168.68.114	103.230.106.216	HTTP	408	GET /time.php HTTP/1.1
10112	197.835697	103.230.106.216	192.168.68.114	HTTP	453	HTTP/1.1 200 OK (text/html)
10124	198.831279	192.168.68.114	103.230.106.216	HTTP	408	GET /time.php HTTP/1.1
10126	198.836827	103.230.106.216	192.168.68.114	HTTP	453	HTTP/1.1 200 OK (text/html)
10141	199.126568	192.168.68.114	103.230.106.216	HTTP	408	GET /time.php HTTP/1.1
10144	199.132840	103.230.106.216	192.168.68.114	HTTP	453	HTTP/1.1 200 OK (text/html)
10173	199.862532	192.168.68.114	103.230.106.216	HTTP	408	GET /time.php HTTP/1.1
10175	199.868314	103.230.106.216	192.168.68.114	HTTP	453	HTTP/1.1 200 OK (text/html)
10185	200.358450	192.168.68.114	103.230.106.216	HTTP	408	GET /time.php HTTP/1.1
10195	200.370827	103.230.106.216	192.168.68.114	HTTP	453	HTTP/1.1 200 OK (text/html)
10261	201.355030	192.168.68.114	103.230.106.216	HTTP	408	GET /time.php HTTP/1.1
10266	201.366735	103.230.106.216	192.168.68.114	HTTP	453	HTTP/1.1 200 OK (text/html)
10267	201.367926	192.168.68.114	103.230.106.216	HTTP	408	GET /time.php HTTP/1.1
10271	201.373886	103.230.106.216	192.168.68.114	HTTP	453	HTTP/1.1 200 OK (text/html)

http.request

A frame operates at the Network Interface Layer of the TCP/IP model and in the OSI model, frames operate at the Data Link Layer (Layer 2). It encapsulates packets by adding MAC addresses and error-checking codes. It enables direct communication between devices on the same local network segment. Frames are the data units transmitted across the physical network medium like Ethernet or Wi-Fi.

```
▼ Frame 2591: Packet, 489 bytes on wire (3912 bits), 489 bytes captured (3912 bits) on interface \Device\NPF_{A052A750-8239-4804-BAA9-7A4E6ED9B85D}, id 0
  Section number: 1
  ▼ Interface id: 0 (\Device\NPF_{A052A750-8239-4804-BAA9-7A4E6ED9B85D})
    Interface name: \Device\NPF_{A052A750-8239-4804-BAA9-7A4E6ED9B85D}
    Interface description: Wi-Fi
    Encapsulation type: Ethernet (1)
    Arrival Time: Nov 12, 2025 18:14:57.936259000 Bangladesh Standard Time
    UTC Arrival Time: Nov 12, 2025 12:14:57.936259000 UTC
    Epoch Arrival Time: 1762949697.936259000
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 1.370000 milliseconds]
    [Time since reference or first frame: 21.909519000 seconds]
    Frame Number: 2591
    Frame Length: 489 bytes (3912 bits)
    Capture Length: 489 bytes (3912 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:ip:tcp:http]
    Character encoding: ASCII (0)
    [Coloring Rule Name: HTTP]
    [Coloring Rule String: http || tcp.port == 80 || http2]
```

Ethernet is a widely used networking technology that operates at the Data Link Layer (Layer 2) and defines how devices communicate over wired local area networks (LANs). It uses MAC addresses to identify devices and organize data into frames for reliable transmission between computers, switches, and other network equipment.

```
▼ Ethernet II, Src: Intel_61:94:05 (dc:21:48:61:94:05), Dst: TpLinkTechno_fd:c0:4c (c0:c9:e3:fd:c0:4c)
  ▼ Destination: TpLinkTechno_fd:c0:4c (c0:c9:e3:fd:c0:4c)
    .... ..0. .... = LG bit: Globally unique address (factory default)
    .... ..0. .... = IG bit: Individual address (unicast)
  ▼ Source: Intel_61:94:05 (dc:21:48:61:94:05)
    .... ..0. .... = LG bit: Globally unique address (factory default)
    .... ..0. .... = IG bit: Individual address (unicast)
  Type: IPv4 (0x0800)
  [Stream index: 0]
```

IPv4 (Internet Protocol version 4) is the core protocol of the Network layer in the TCP/IP model, responsible for addressing and routing data packets across networks. It uses 32-bit addresses, typically written in dot-decimal format (e.g., 192.168.0.1). IPv4 uses source and destination IP addresses to identify the sender and receiver. For example, in this traffic, the source IP address is 192.168.68.114 and the destination IP address is 103.230.106.216. It supports about 4.3 billion unique addresses. IPv4 packets include fields for source and destination addresses, header checksum and time-to-live (TTL) to manage delivery and reliability. Due to the exhaustion of available IPv4 addresses, it is gradually being replaced by IPv6, which offers a much larger address space.

```
▼ Internet Protocol Version 4, Src: 192.168.68.114, Dst: 103.230.106.216
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  ▼ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    0000 00.. = Differentiated Services Codepoint: Default (0)
    .... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
  Total Length: 475
  Identification: 0x19cc (6604)
  ▼ 010. .... = Flags: 0x2, Don't fragment
    0... .... = Reserved bit: Not set
    .1.. .... = Don't fragment: Set
    ..0. .... = More fragments: Not set
    ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 128
  Protocol: TCP (6)
  Header Checksum: 0x0000 [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 192.168.68.114
  Destination Address: 103.230.106.216
  [Stream index: 32]
```

TCP (Transmission Control Protocol) is a core transport layer protocol in the TCP/IP model that ensures reliable, ordered, and error-checked delivery of data between applications. It establishes a connection-oriented communication channel using a three-way handshake and manages flow control, congestion control, and retransmission of lost packets to maintain data integrity.

Here, Source Port : 61449, Destination Port: 80, Seq: 1, Ack: 1

```
▼ Transmission Control Protocol, Src Port: 61449, Dst Port: 80, Seq: 1, Ack: 1, Len: 435
  Source Port: 61449
  Destination Port: 80
  [Stream index: 18]
  [Stream Packet Number: 4]
  ▼ [Conversation completeness: Incomplete, DATA (15)]
    ..0. .... = RST: Absent
    ...0 .... = FIN: Absent
    .... 1... = Data: Present
    .... .1.. = ACK: Present
    .... ..1. = SYN-ACK: Present
    .... ...1 = SYN: Present
    [Completeness Flags: ..DASS]
  [TCP Segment Len: 435]
  Sequence Number: 1      (relative sequence number)
  Sequence Number (raw): 2822851784
  [Next Sequence Number: 436      (relative sequence number)]
  Acknowledgment Number: 1      (relative ack number)
  Acknowledgment number (raw): 1272830455
  0101 .... = Header Length: 20 bytes (5)
  ▼ Flags: 0x018 (PSH, ACK)
    000. .... = Reserved: Not set
    ...0 .... = Accurate ECN: Not set
    .... 0... = Congestion Window Reduced: Not set
    .... .0.. = ECN-Echo: Not set
    .... ..0. = Urgent: Not set
    .... ...1 = Acknowledgment: Set
    .... .... 1... = Push: Set
    .... .... .0.. = Reset: Not set
    .... .... ..0. = Syn: Not set
    .... .... ...0 = Fin: Not set
    [TCP Flags: .....AP...]
  Window: 65535
  [Calculated window size: 65535]
  [Window size scaling factor: -2 (no window scaling used)]
  Checksum: 0xd9a6 [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
```

```
▼ [Timestamps]
  [Time since first frame in this TCP stream: 43.303000 milliseconds]
  [Time since previous frame in this TCP stream: 38.194000 milliseconds]
  ▼ [SEQ/ACK analysis]
    [iRTT: 5.109000 milliseconds]
    [Bytes in flight: 435]
    [Bytes sent since last PSH flag: 435]
  [Client Contiguous Streams: 1]
  [Server Contiguous Streams: 1]
  TCP payload (435 bytes)
```

Hypertext Transfer Protocol (HTTP) is an application layer protocol used for transferring web pages and data between a web server and a client (usually a browser). It defines how messages are formatted, transmitted, and responded to, enabling users to access and interact with websites over the Internet. HTTP (Hypertext Transfer Protocol) is not secure, as data is sent in plain text and can be intercepted. Its secure version, HTTPS which encrypts data using SSL/TLS, ensures privacy and data integrity during transmission.

```
▼ Hypertext Transfer Protocol
  ▼ GET / HTTP/1.1\r\n
    Request Method: GET
    Request URI: /
    Request Version: HTTP/1.1
    Host: cspb.teletalk.com.bd\r\n
    Connection: keep-alive\r\n
    Upgrade-Insecure-Requests: 1\r\n
    User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/142.0.0.0 Safari/537.36\r\n
    Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7\r\n
    Accept-Encoding: gzip, deflate\r\n
    Accept-Language: en-US,en;q=0.9\r\n
    \r\n
    [Full request URI: http://cspb.teletalk.com.bd/]
```

http.response

The Frame falls under Network Interface Layer in the TCP/IP model and Data Link layer in the OSI model handles framing, error detection, and physical addressing of data for transmission over the network. It ensures reliable delivery of packets between directly connected devices using MAC addresses.

```
▼ Frame 2718: Packet, 179 bytes on wire (1432 bits), 179 bytes captured (1432 bits) on interface \Device\NPF_{A052A750-8239-4804-BAA9-7A4E6ED9B85D}, id 0
  Section number: 1
  ▼ Interface id: 0 (\Device\NPF_{A052A750-8239-4804-BAA9-7A4E6ED9B85D})
    Interface name: \Device\NPF_{A052A750-8239-4804-BAA9-7A4E6ED9B85D}
    Interface description: Wi-Fi
    Encapsulation type: Ethernet (1)
    Arrival Time: Nov 12, 2025 18:14:58.023514000 Bangladesh Standard Time
    UTC Arrival Time: Nov 12, 2025 12:14:58.023514000 UTC
    Epoch Arrival Time: 1762949698.023514000
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 0.000000000 seconds]
    [Time since reference or first frame: 21.996774000 seconds]
    Frame Number: 2718
    Frame Length: 179 bytes (1432 bits)
    Capture Length: 179 bytes (1432 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:ip:tcp:http:png]
    Character encoding: ASCII (0)
    [Coloring Rule Name: HTTP]
    [Coloring Rule String: http || tcp.port == 80 || http2]
```

Ethernet is a widely used network technology that connects devices in a local area network (LAN) using wired connections. It defines standards for data transmission, frame structure, and error handling, ensuring efficient and reliable communication between devices

```
▼ Ethernet II, Src: TplinkTechno_fd:c0:4c (c0:c9:e3:fd:c0:4c), Dst: Intel_61:94:05 (dc:21:48:61:94:05)
  ▼ Destination: Intel_61:94:05 (dc:21:48:61:94:05)
    .... ..0. .... = LG bit: Globally unique address (factory default)
    .... ...0 .... = IG bit: Individual address (unicast)
  ▼ Source: TplinkTechno_fd:c0:4c (c0:c9:e3:fd:c0:4c)
    .... ..0. .... = LG bit: Globally unique address (factory default)
    .... ...0 .... = IG bit: Individual address (unicast)
  Type: IPv4 (0x0800)
  [Stream index: 0]
```

IPv4 (Internet Protocol version 4) operates at the Network Layer (Layer 3) of the OSI model and is responsible for addressing and routing packets across different networks. It uses 32-bit addresses written in dotted-decimal notation like 192.168.1.1. It provides approximately 4.3 billion unique addresses. There is a source IP and destination IP. Here for source IP address, we get 103.230.106.216 and for destination IP address, we get 192.168.68.114. It has features like packet fragmentation, time-to-live values to prevent infinite loops and header checksums for error detection during data transmission across the internet.

```
▼ Internet Protocol Version 4, Src: 103.230.106.216, Dst: 192.168.68.114
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  ▼ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    0000 00.. = Differentiated Services Codepoint: Default (0)
    .... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
  Total Length: 165
  Identification: 0xca90 (51856)
  ▼ 010. .... = Flags: 0x2, Don't fragment
    0... .... = Reserved bit: Not set
    .1.. .... = Don't fragment: Set
    ..0. .... = More fragments: Not set
    ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 243
  Protocol: TCP (6)
  Header Checksum: 0xe4e8 [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 103.230.106.216
  Destination Address: 192.168.68.114
  [Stream index: 32]
```

TCP (Transmission Control Protocol) operates at the Transport Layer (Layer 4) of the OSI model and provides reliable, connection-oriented communication between applications over a network. It establishes connections through a three-way handshake, guarantees data delivery by acknowledging received segments and retransmitting lost ones, maintains proper sequencing of data packets, and controls data flow to prevent network congestion and ensure information arrives complete and in the correct order. Here, Source Port: 80, Destination Port: 61601, Seq: 54021, Ack: 394

▼ Transmission Control Protocol, Src Port: 80, Dst Port: 61601, Seq: 54021, Ack: 394, Len: 125

Source Port: 80

Destination Port: 61601

[Stream index: 17]

[Stream Packet Number: 44]

▼ [Conversation completeness: Incomplete, DATA (15)]

..0. = RST: Absent

...0 = FIN: Absent

.... 1... = Data: Present

.... .1.. = ACK: Present

.... ..1. = SYN-ACK: Present

.... ...1 = SYN: Present

[Completeness Flags: ..DASS]

[TCP Segment Len: 125]

Sequence Number: 54021 (relative sequence number)

Sequence Number (raw): 4252270079

[Next Sequence Number: 54146 (relative sequence number)]

Acknowledgment Number: 394 (relative ack number)

Acknowledgment number (raw): 127772110

0101 = Header Length: 20 bytes (5)

▼ Flags: 0x018 (PSH, ACK)

000. = Reserved: Not set

...0 = Accurate ECN: Not set

.... 0... = Congestion Window Reduced: Not set

.... .0.. = ECN-Echo: Not set

.... ..0. = Urgent: Not set

.... ...1 = Acknowledgment: Set

.... 1... = Push: Set

....0.. = Reset: Not set

....0. = Syn: Not set

....0 = Fin: Not set

[TCP Flags:AP...]

Window: 4773

[Calculated window size: 4773]

[Window size scaling factor: -2 (no window scaling used)]

Checksum: 0xc7a1 [unverified]

[Checksum Status: Unverified]

Urgent Pointer: 0

▼ [Timestamps]

[Time since first frame in this TCP stream: 130.705000 milliseconds]

[Time since previous frame in this TCP stream: 0.000000000 seconds]

[Client Contiguous Streams: 1]

[Server Contiguous Streams: 1]

TCP payload (125 bytes)

TCP segment data (125 bytes)

HTTP (Hypertext Transfer Protocol) operates at the Application Layer (Layer 7) and defines how web browsers and servers communicate to exchange web pages and resources. It uses request-response methods like GET and POST over TCP port 80, but transmits data in plain text without encryption, making it insecure. HTTPS (HTTP Secure) is the secure version that encrypts all data using SSL/TLS protocols to protect information from interception during transmission.

```
▼ Hypertext Transfer Protocol
  ▼ HTTP/1.1 200 OK\r\n
    Response Version: HTTP/1.1
    Status Code: 200
    [Status Code Description: OK]
    Response Phrase: OK
    Server: nginx\r\n
    Date: Sun, 16 Nov 2025 12:32:00 GMT\r\n
    Content-Type: image/png\r\n
  ▼ Content-Length: 53777\r\n
    [Content length: 53777]
    Last-Modified: Mon, 01 Jun 2020 09:40:06 GMT\r\n
    Connection: keep-alive\r\n
    ETag: "5ed4ccf6-d211"\r\n
    Expires: Tue, 16 Dec 2025 12:32:00 GMT\r\n
    Cache-Control: max-age=2592000\r\n
    Cache-Control: public\r\n
    Pragma: public\r\n
    Vary: Accept_encoding\r\n
    Accept-Ranges: bytes\r\n
    \r\n
    [Request in frame: 17493]
    [Time since request: 24.628000 milliseconds]
    [Request URI: /images/govt_logo.png]
    [Full request URI: http://cspb.teletalk.com.bd/images/govt_logo.png]
    File Data: 53777 bytes
```